DIGITAL IMAGE PROCESSING (DIP)

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Section 3: Image Histogram

Fall 1405
Fall 2026

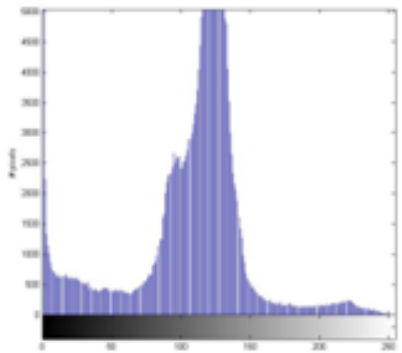
Outlier

In all parts, we will use the definitions in MATLAB.
Students can also use Python software.

- Different Techniques in Histogram Enhancement
- Histogram Equalization
- Histogram Matching

What does histogram mean?





The function h(k) determines the number of pixels with gray level k in an image with size M×N.

$$\rho_k(k) = \frac{h(k)}{MN};\quad \sum_{k=0}^{L-1} \rho_k(k) = 1$$

$\rho_k(k)$ would be

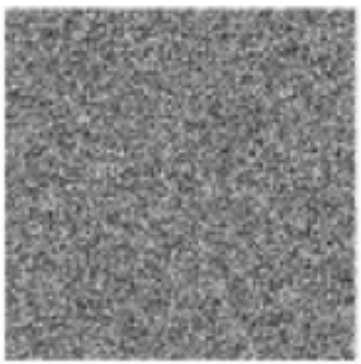
Probability Mass Functiom (PMF)

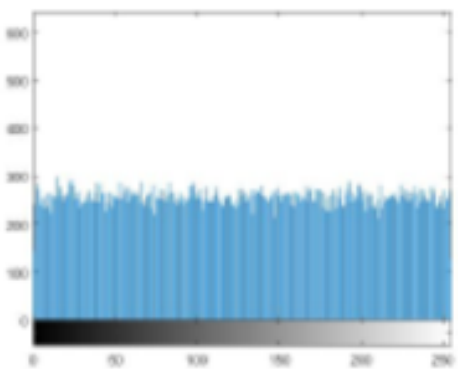
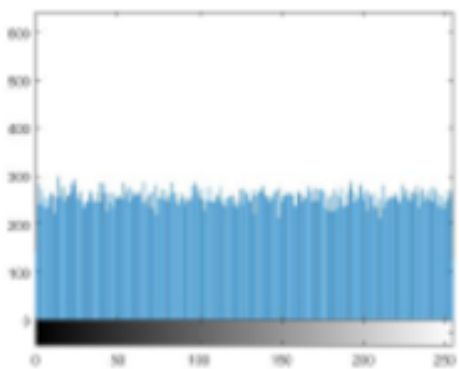
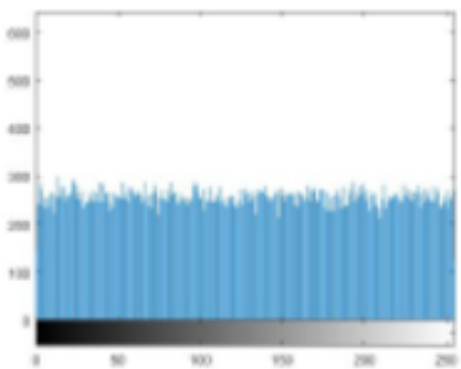
Matlab Commands: imhist, histogram

Histogram Examples:

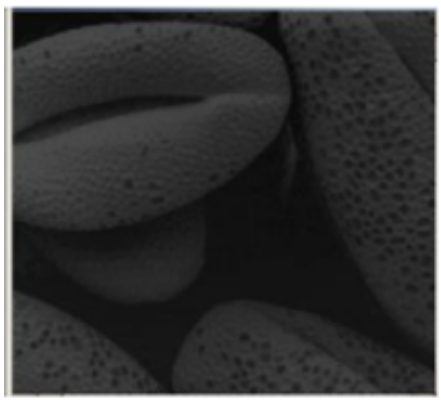
Histograms illustrate the distribution of individual pixel values in an image. The histogram demonstrates strong robustness to changes in pixel position and structural variations.

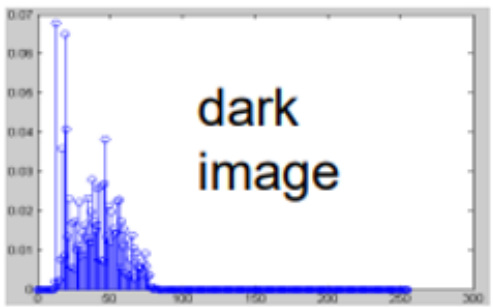
There could be different images with the same histogram distributions.

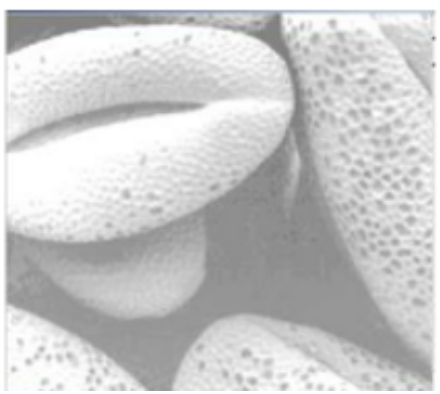


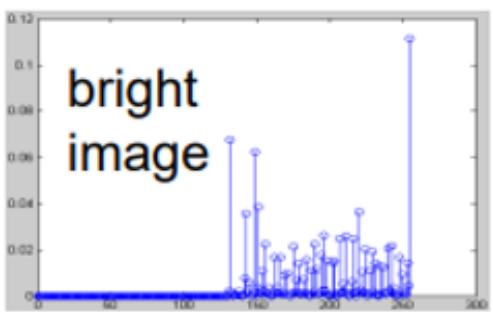





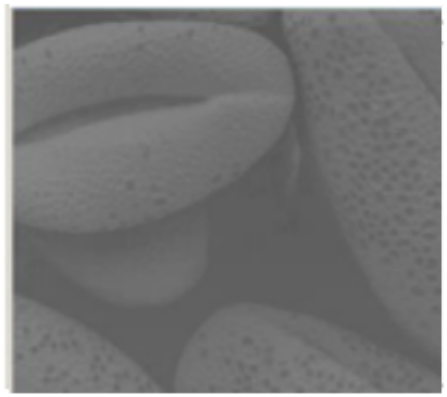
Histogram Examples

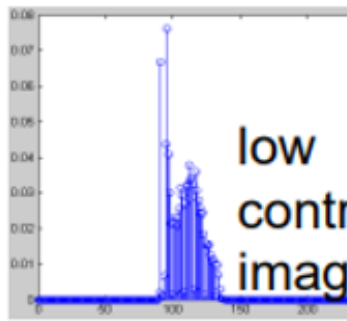


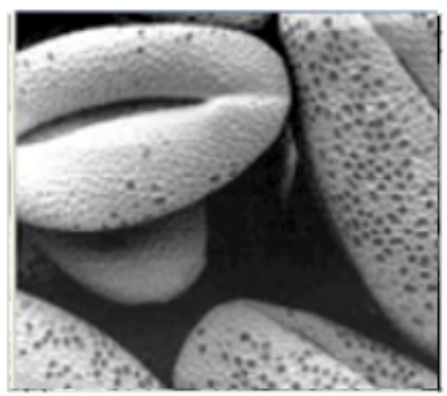


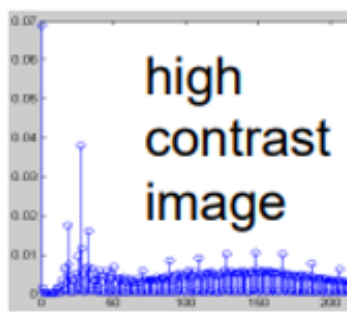












Point Operations methods: $g(m,n) = T(f(m,n))$:

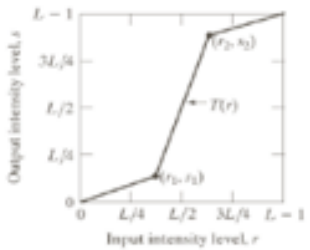
Point operation methods or intensity transformations of the form: let's $f(m,n) = r; \quad s = T[r]; \quad g(m,n) = s; \quad 0 \leq r, s \leq max(brightness)$
r and s are the grey levels in $(m,n)^{th}$ pixel in two images.
A few techniques are:

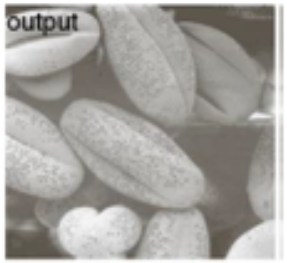
- Contrast manipulation (enhancement/stretching/threshold)
- Image negatives
- Log transformations
- Power log transformations
- Grey-level slicing
- Bit-plane slicing
- Histogram processing
- Others

Examples:

Contrast Stretching





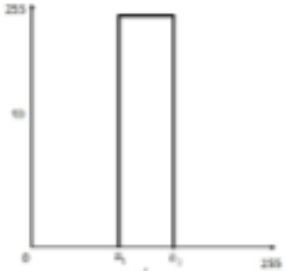


Thresholding



Slicing

Remove background

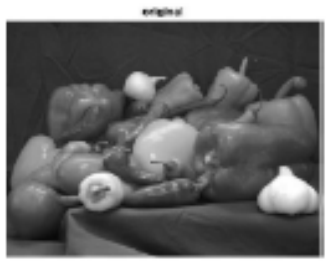




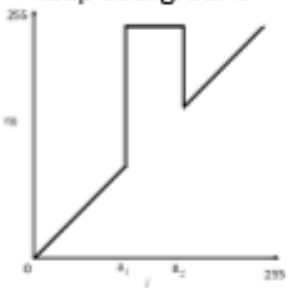




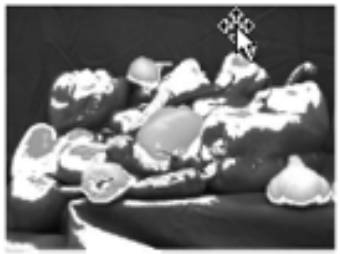
Values [50,100] [100,150] [200:255]

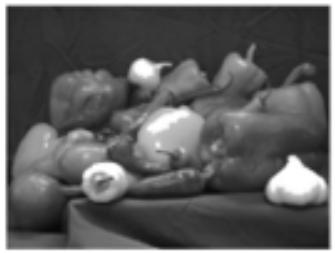


Keep background



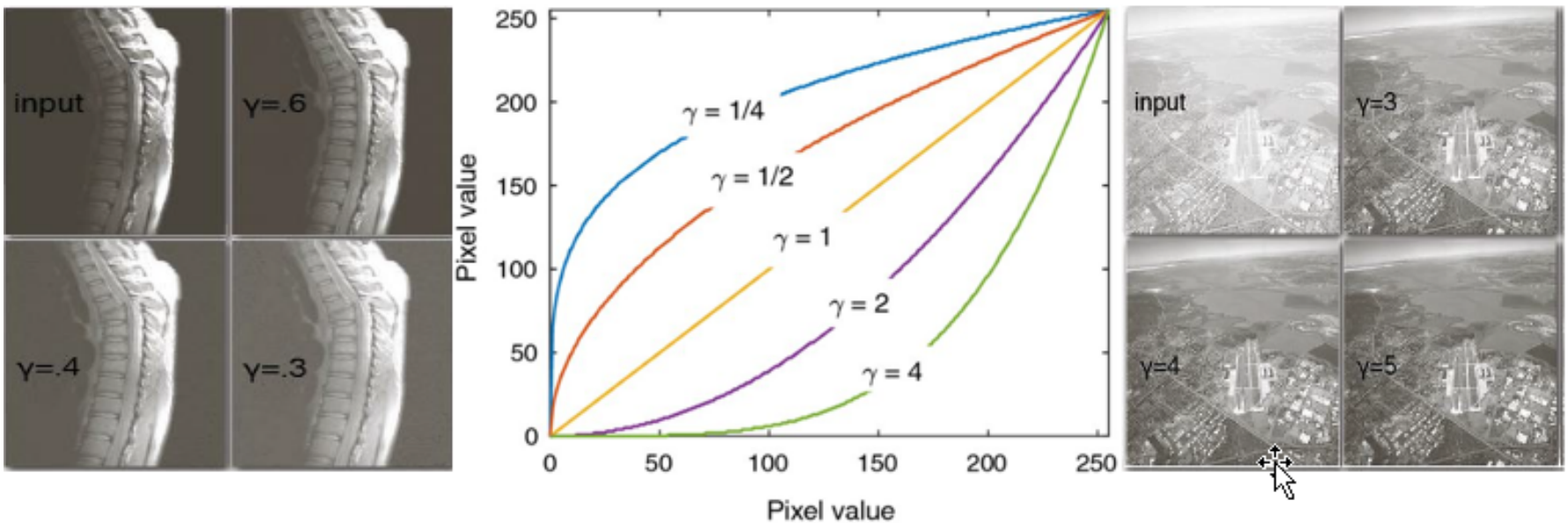






Power Law Transform

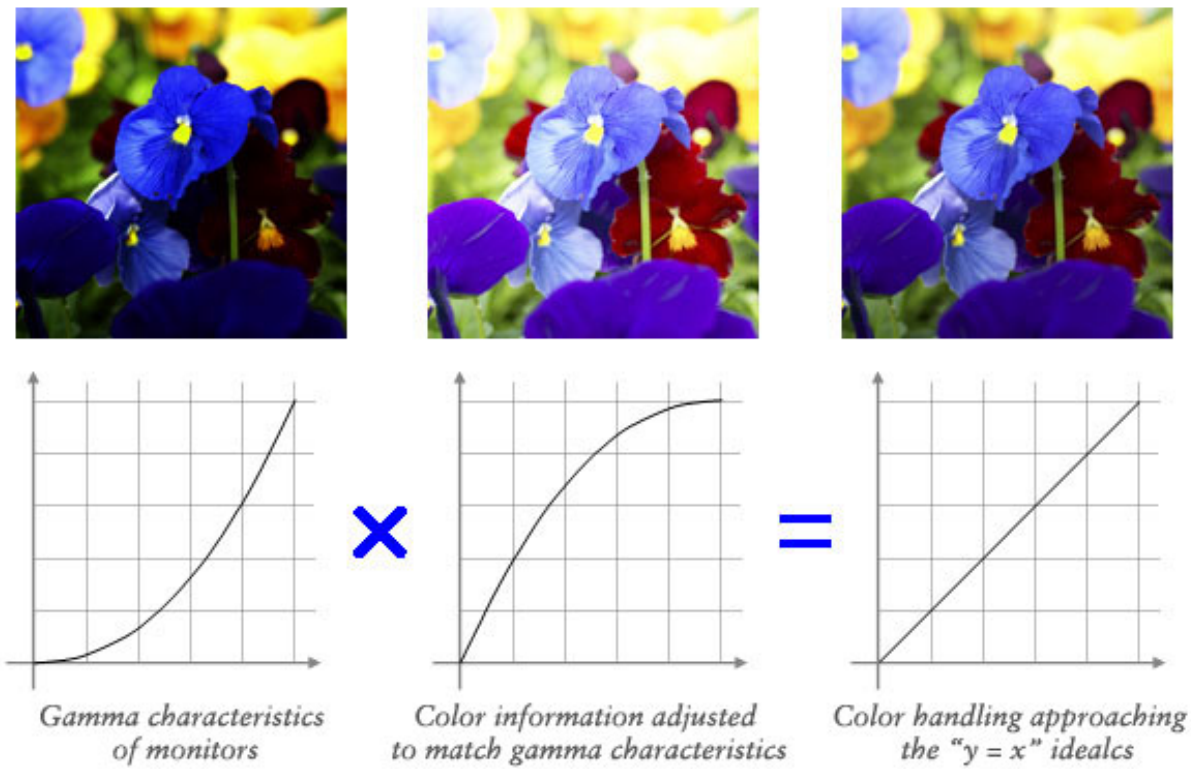
$S = Cr^{\gamma}$: C and γ are fix values, C is used to normalize intensity range.



What is difference between Brightness range $[0 - 1]$ and $[0 - 255]$?

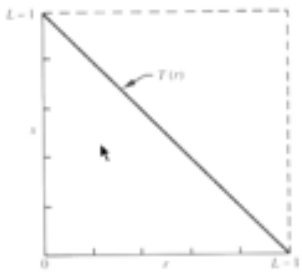
Gamma Correction

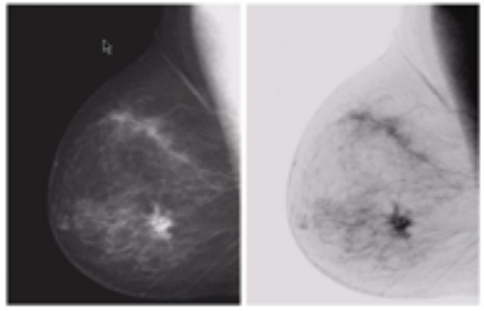
Ordinarily, the nature of monitor gamma is such that intermediate tones tend to appear dark. Efforts seek to promote accurate exchange of color information by inputting data signals in which the intermediate tones have already been brightened to approach an input:output balance of 1:1. Balancing color information to match device gamma characteristics in this way is called gamma correction.



Other transforms

Image negatives



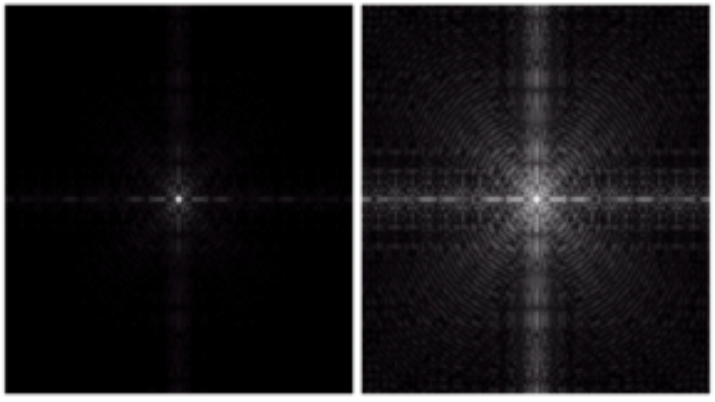


a b
FIGURE 3.4
(a) Original digital mammogram. (b) Negative image obtained using the negative transformation in Eq. (3.2-1). (Courtesy of G.E. Medical Systems.)

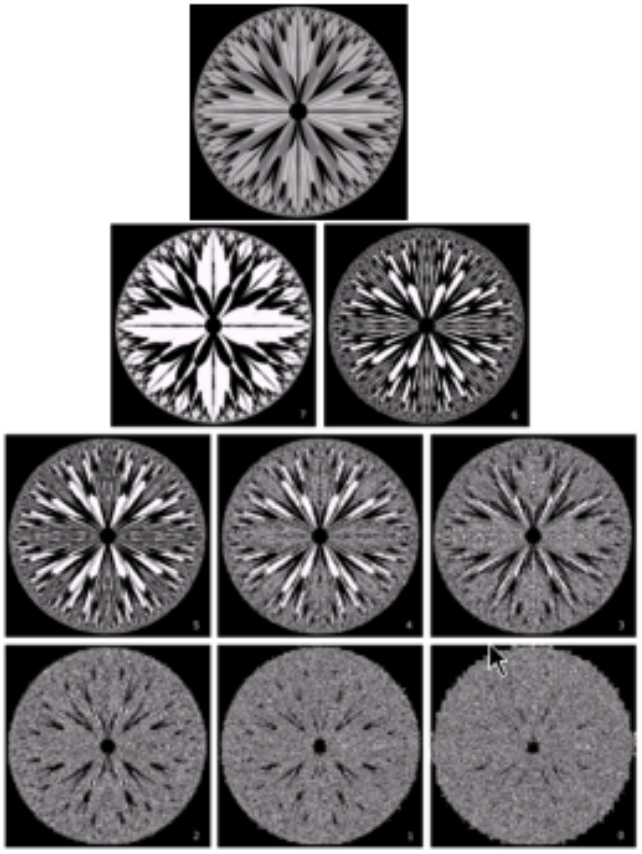
Log Transform

a b
FIGURE 3.5
(a) Fourier spectrum. (b) Result of applying the log transformation given in Eq. (3.2-2) with $c = 1$.





Bit Slicing



Thresholding in RGB Space

MAP detector in RGB-space

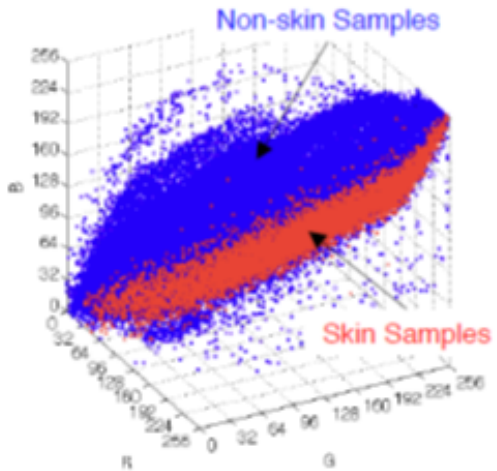


Original image



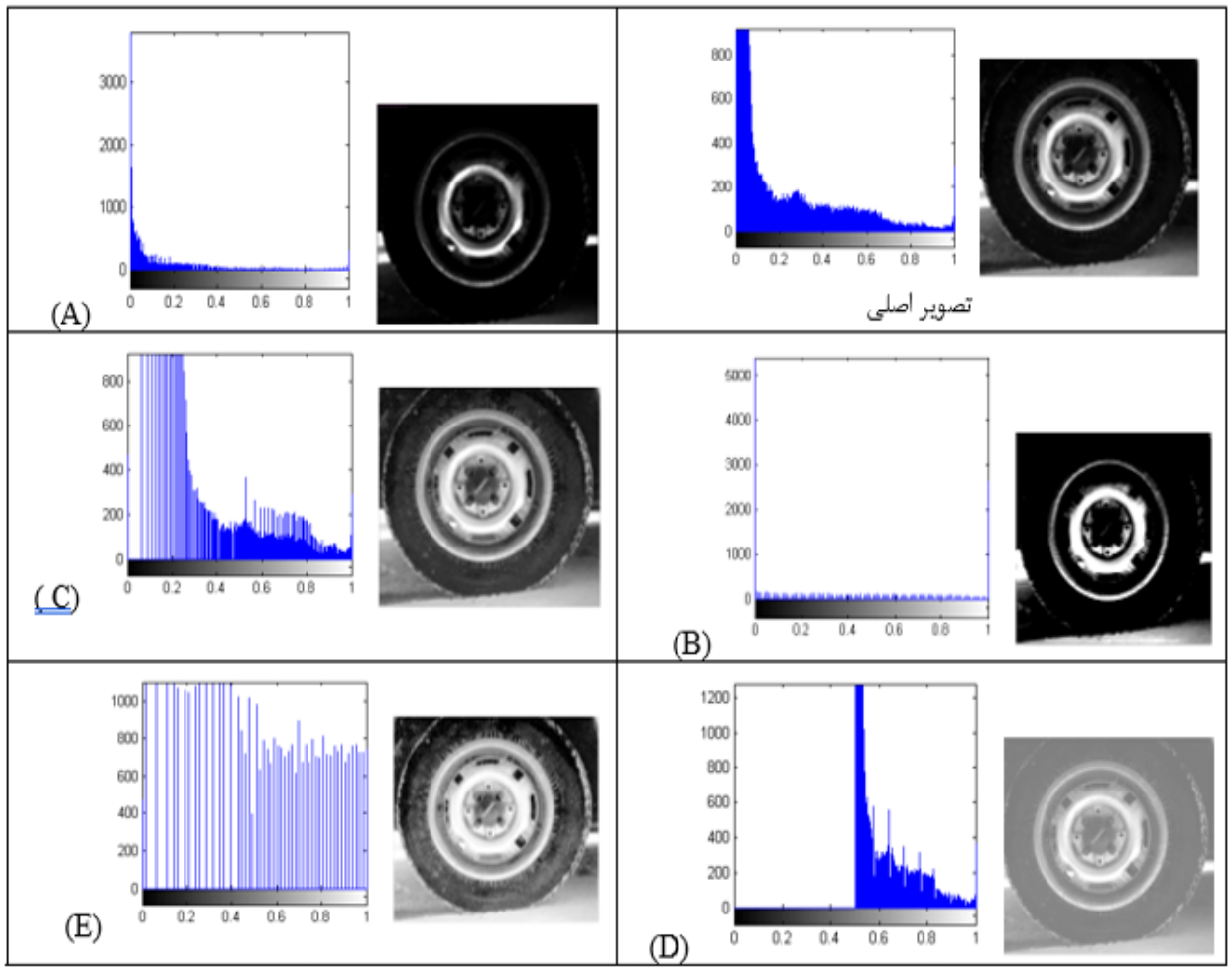
Skin color detector

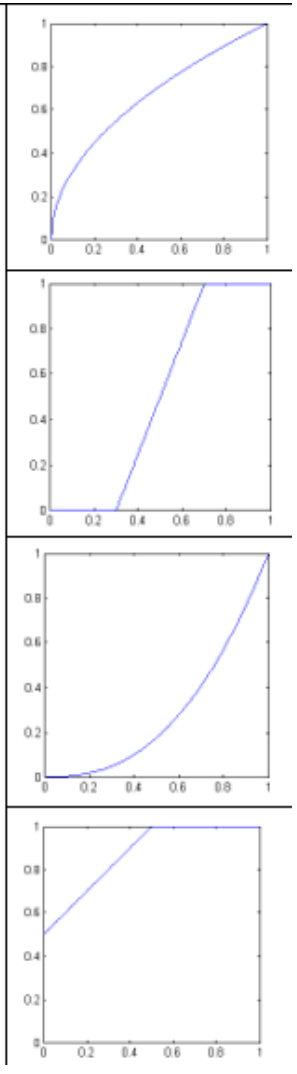
Using a lot of images to train algorithm for skin and non skin points



Test:

Which Transform leads to which image?



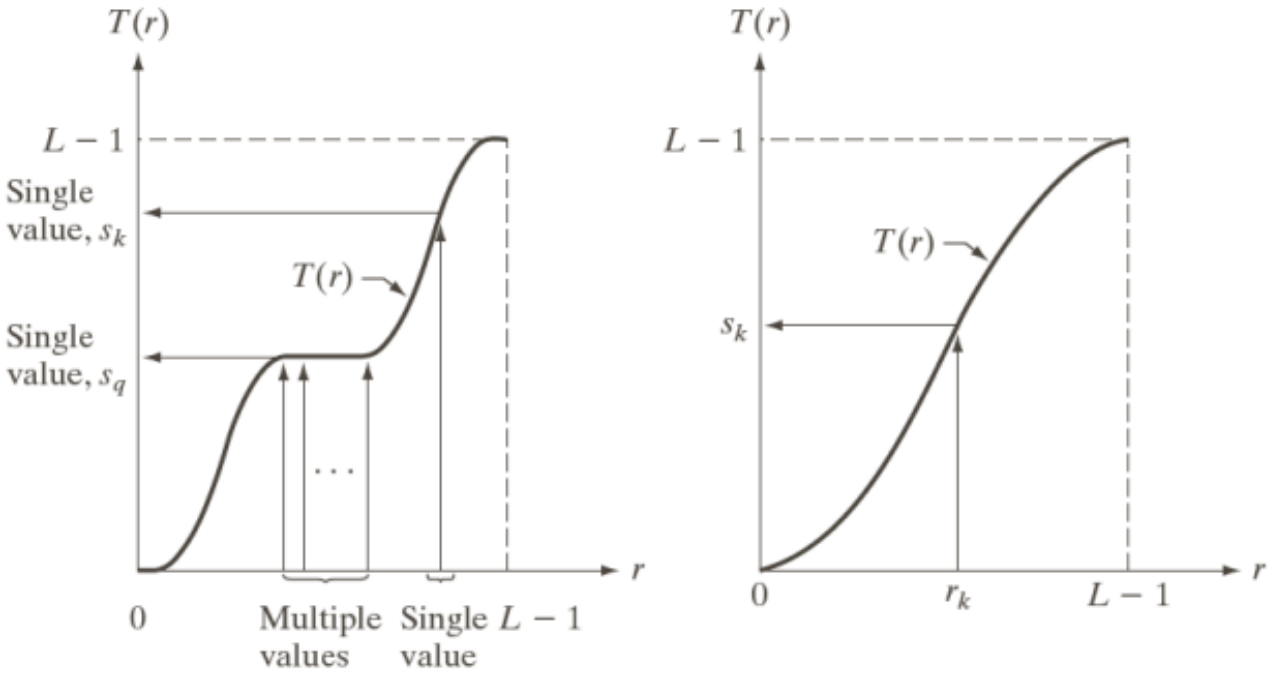


Intensity Transform Constraints to Enhance Presentation:

Consider grey scale images, all Transforms:
$s = T(r)$

- value ranges of r and s should be the same.
- Transform should be monotonically increasing, so guarantees that ordering of the output intensity values will follow the ordering of the input intensity values;
  $g(m,n) = r_1 \leq g(i,j) = r_2$ iff
  $s_1 = T(r_1) = f(m,n) < s_2 = T(r_2) = f(i,j)$
- If and only if transform be strictly increasing, transform would be reversible.

Monotonicity versus Strict Monotonicity



- In the above Figure on the left we cannot perform inverse mapping (from s to r). This is because there is a case where a single value s_q is the mapping of multiple original intensities.
- In the above Figure on the right inverse mapping is possible.

Modelling Histogram: Histogram Equalization

Objective: assign a new value to each brightness level so that the output image histogram approaches a flat distribution of shades of grey.

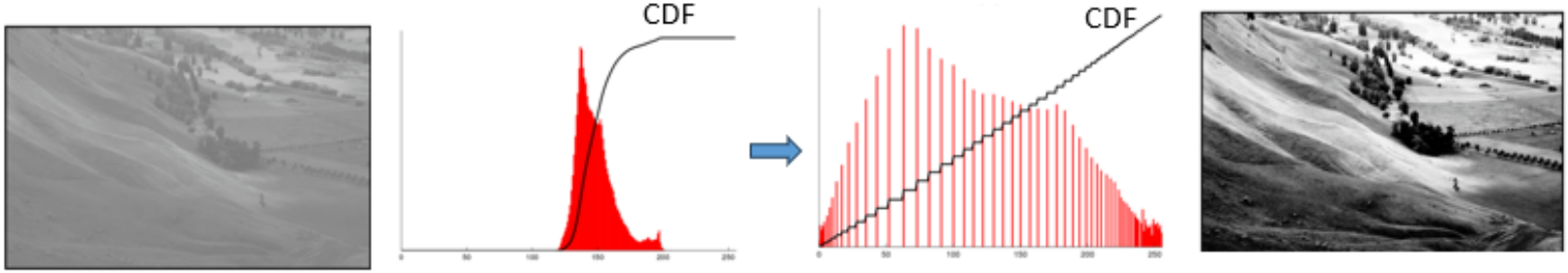
Let in continuous space: $s = T(r) \rightarrow$

Based on probability theory: $\rho_s(s) = \rho_r(r) \left[\frac{\mathrm{d}T(r))}{\mathrm{d}r}\right]^{-1}$

For equalization enhancement $\rho(s)$ should be uniform distribution: $\rho_s(s) = \frac{1}{L-1}$, Therefore:

$s = T(r) = (L-1)\int_0^r \rho_r(w)dw$

what is the mean of $\int_0^r \rho_r(w)dw$



Histogram Equalization Formulas:

For images with discrete gray values,

$$\rho_r(r_k) = \frac{n_k}{n} \quad for\ 0 \le r_k \le L-1$$

L: Total number of gray levels

n_k: Number of pixels with gray value r_k

n: Total number of pixels in the image

The discrete version of the histogram equalization based on CDF is given by:

$$y_k = CDF(\rho_r(r_k)) = \sum_{i=0}^{k} \frac{n_k}{n} = \sum_{i=0}^{k} \rho_r(r_k)$$
$$for\ 0 \le r_k \le L-1$$

Now, it needs to be converted to the gray levels

$$0 \le s_k \le L-1$$

1: $s_k = \lfloor y_k\,(L-1) + 0.5 \rfloor$

$$s_{L-1} = s_{\max} = L-1$$

or

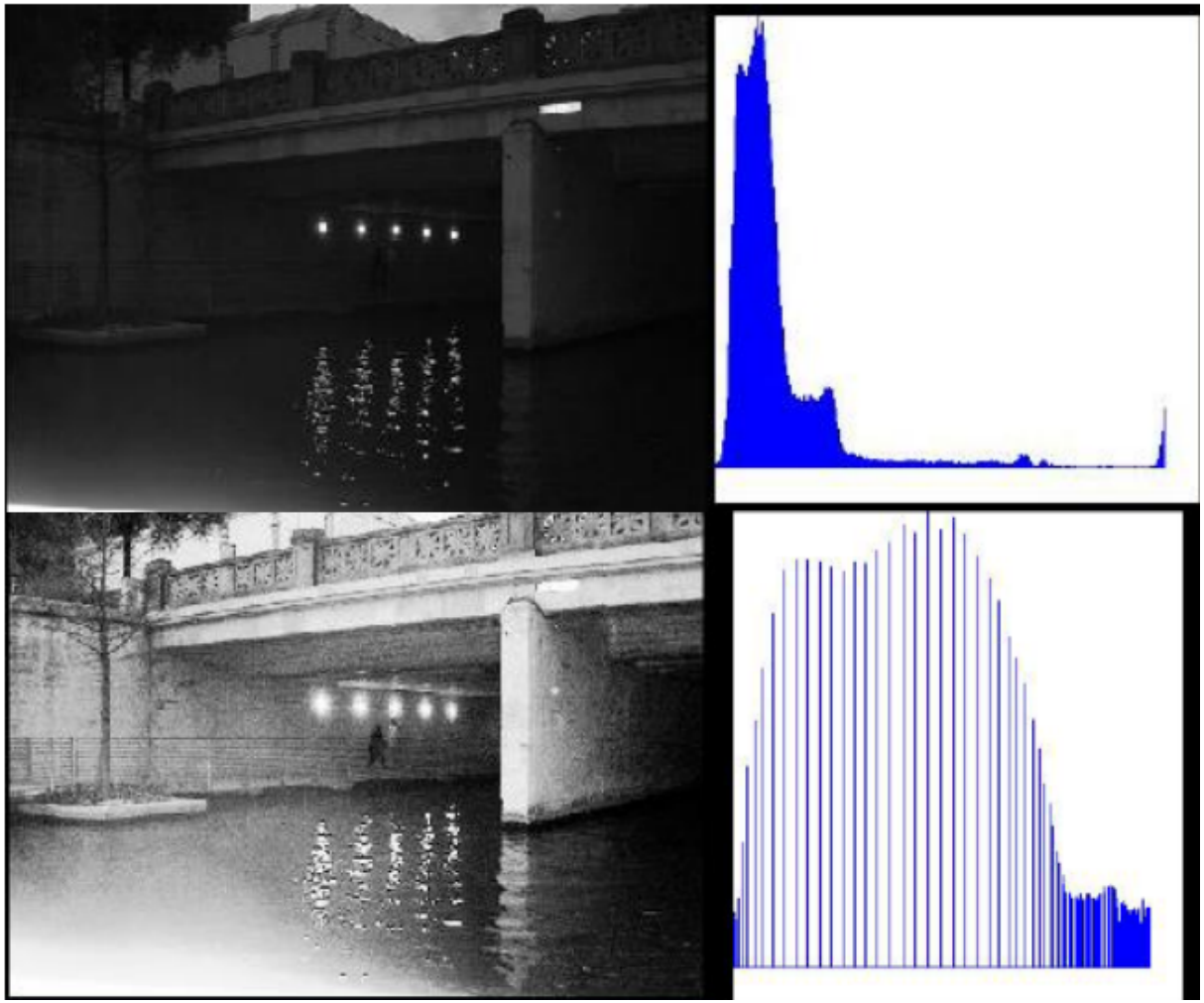
2: $s_k = \left\lfloor \frac{y_k - y_{min}}{1 - y_{min}} (L-1) + 0.5 \right\rfloor$

$$s_0 = 0\ , \qquad s_{L-1} = s_{\max} = L-1$$

or

3: $s_k = round\left[\frac{y_k - y_{min}}{1 - y_{min}} (L-1) \right]$

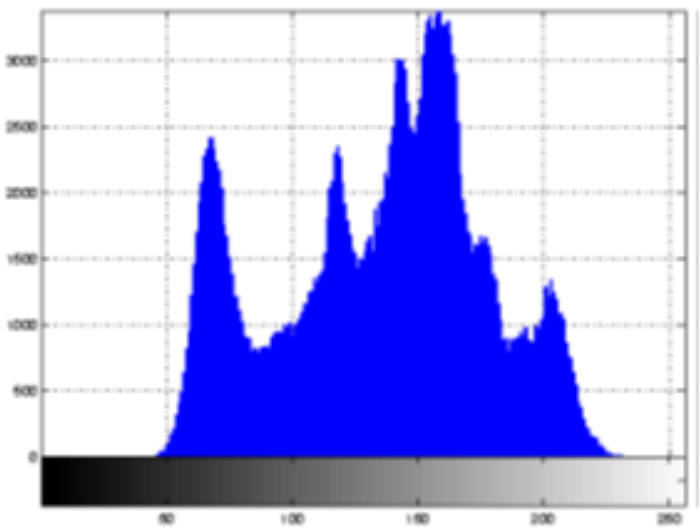
Histogram Equalization: Example



Histogram Equalization: Example

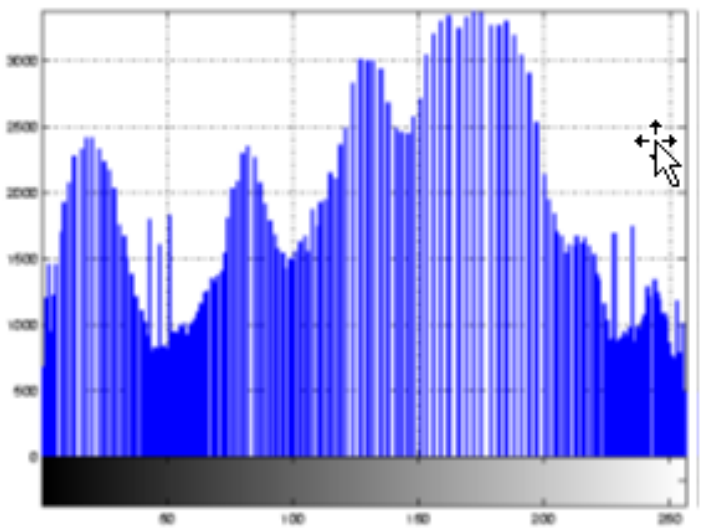
Original image





Equalized image

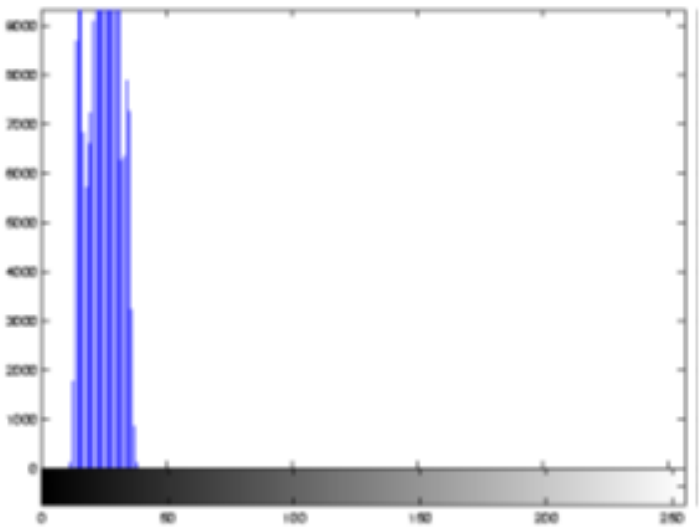


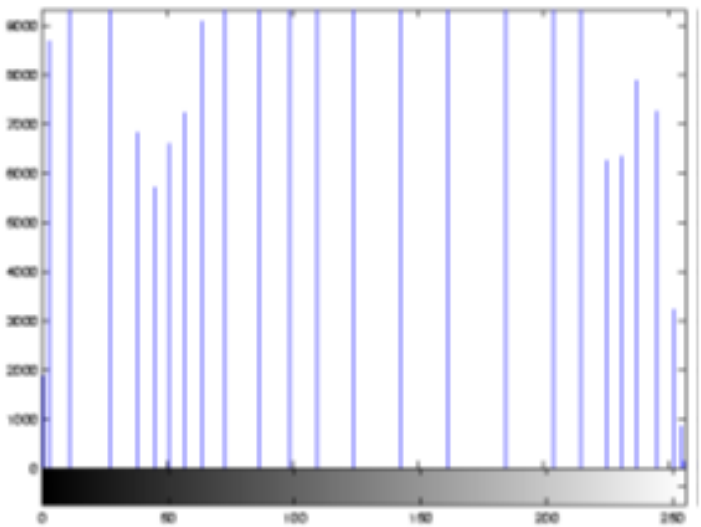


Histogram Equalization: Example





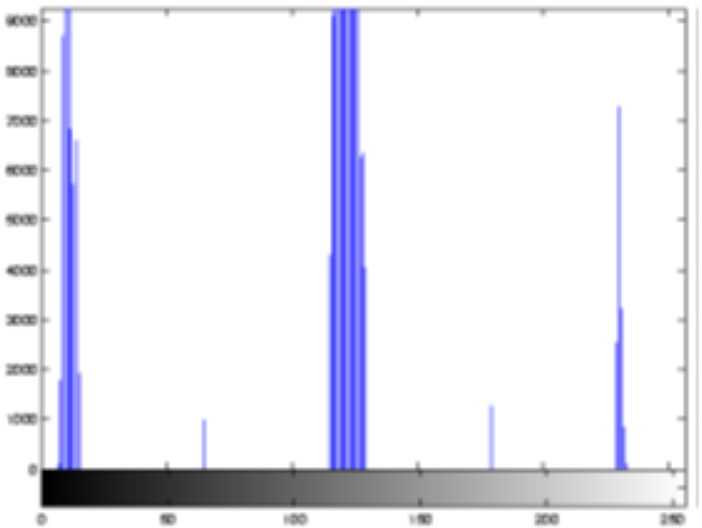


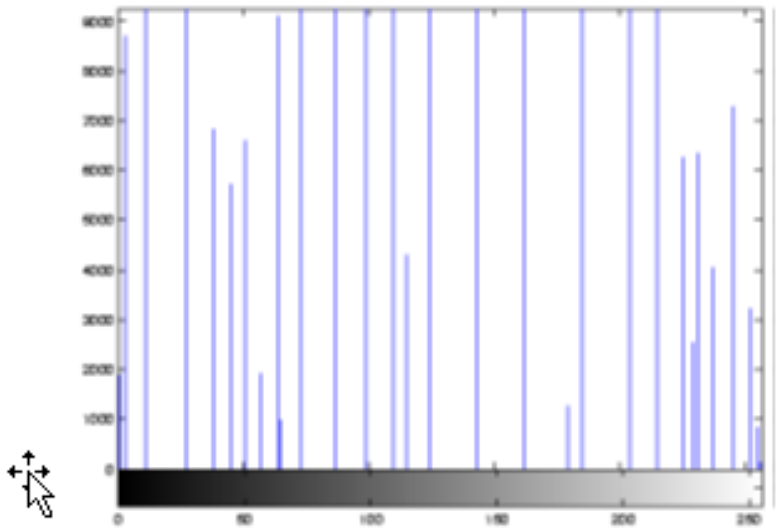


Histogram Equalization: Example



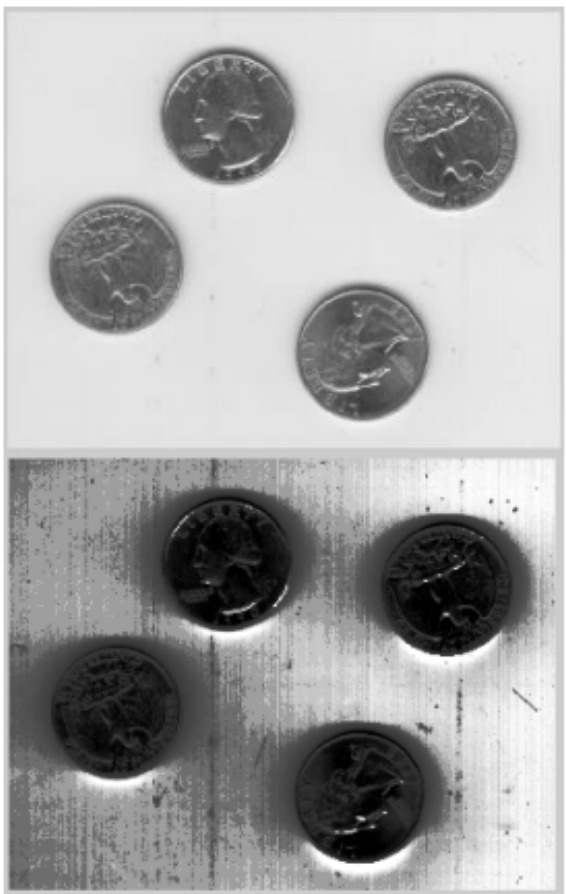


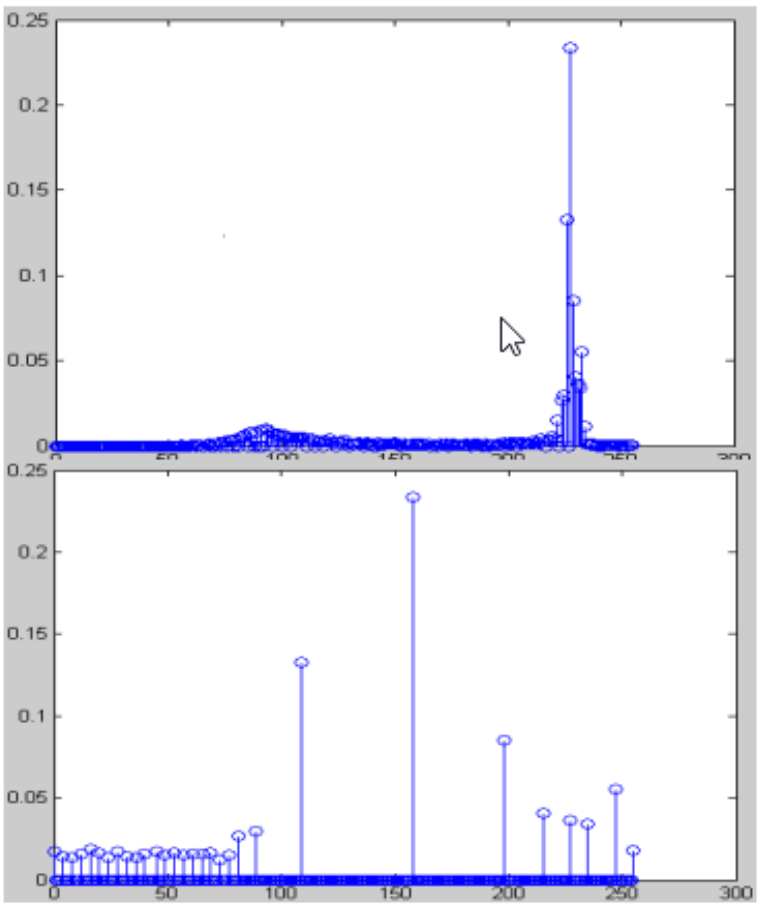




Undesirable Example

Histogram equalization may not always produce desirable results, particularly if the given histogram is very narrow. It can produce false edges and false regions. It can also increase image “graininess” and “patchiness.”



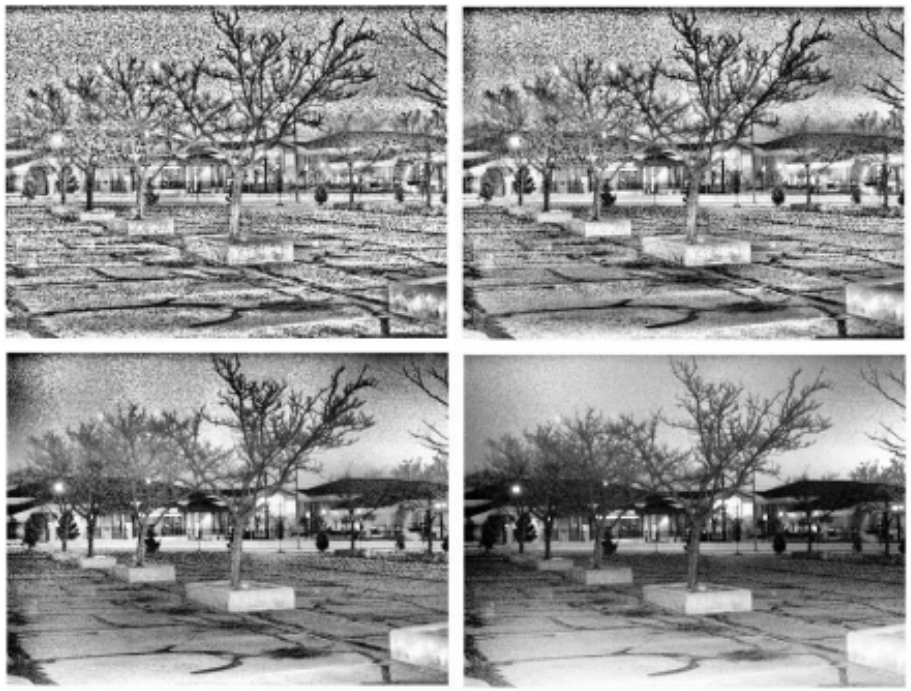


Numerical Example

Example: Assume the images have 64 x 64 = 4096 pixels in 8 gray levels. The following table shows the equalization process corresponding to the two methods:

Pixel value	0	1	2	3	4	5	6	7
Number of pixels	790	1023	850	656	329	245	122	81
Probability	0.19	0.25	0.21	0.16	0.08	0.06	0.03	0.02
Cumulative probability	0.19	0.44	0.65	0.81	0.89	0.95	0.98	1
New Pixel Value (1)	1	3	5	6	6	7	7	7
Number of new pixel values (1)	790	1023	850	985			248	
Probability of new pixel values (1)	0.19	0.25	0.21	0.24			0.11	
New Pixel Value (2)	0	2	4	5	6	7	7	7
Number of new pixel values (2)	790	1023	850	656	329		248	
Probability of new pixel values (2)	0.19	0.25	0.21	0.16	0.08		0.11	

Local Histogram Equalization Example 1



Top left: A window size of [15 15]. Top right: A window size of [30 30].
Bottom left: A window size of [75 75]. Bottom right: A window size of [150 150].

- Small windows enhance local details but may amplify noise.
- Large windows behave more like global histogram equalization.
- Abrupt changes between neighboring regions can create visible artifacts.

Local Histogram Equalization Example 2

Original image
Dental Xray





Global histogram equalization

Adaptive histogram equalization, 8x8 tiles





Adaptive histogram equalization, 16x16 tiles

Contrast-Limited Adaptive Histogram Equalization (CLAHE)(2012)

- Divide the image into small rectangular regions (Tile size).
- Calculate the histogram of each tile.
- Clip histogram bins that exceed a predefined limit (Clip limit).
- Redistribute the excess counts among the bins.
- Perform histogram equalization on each tile.
- interpolation between neighboring tiles to avoid visible boundaries.

Small ClipLimit: More aggressive histogram clipping; less contrast amplification and generally less noise enhancement.
Large ClipLimit: Less clipping; stronger local contrast enhancement, but potentially more noise amplification.

CLAHE Example:

original



Equalized



ClipLimit = 0.001



ClipLimit = 0.01



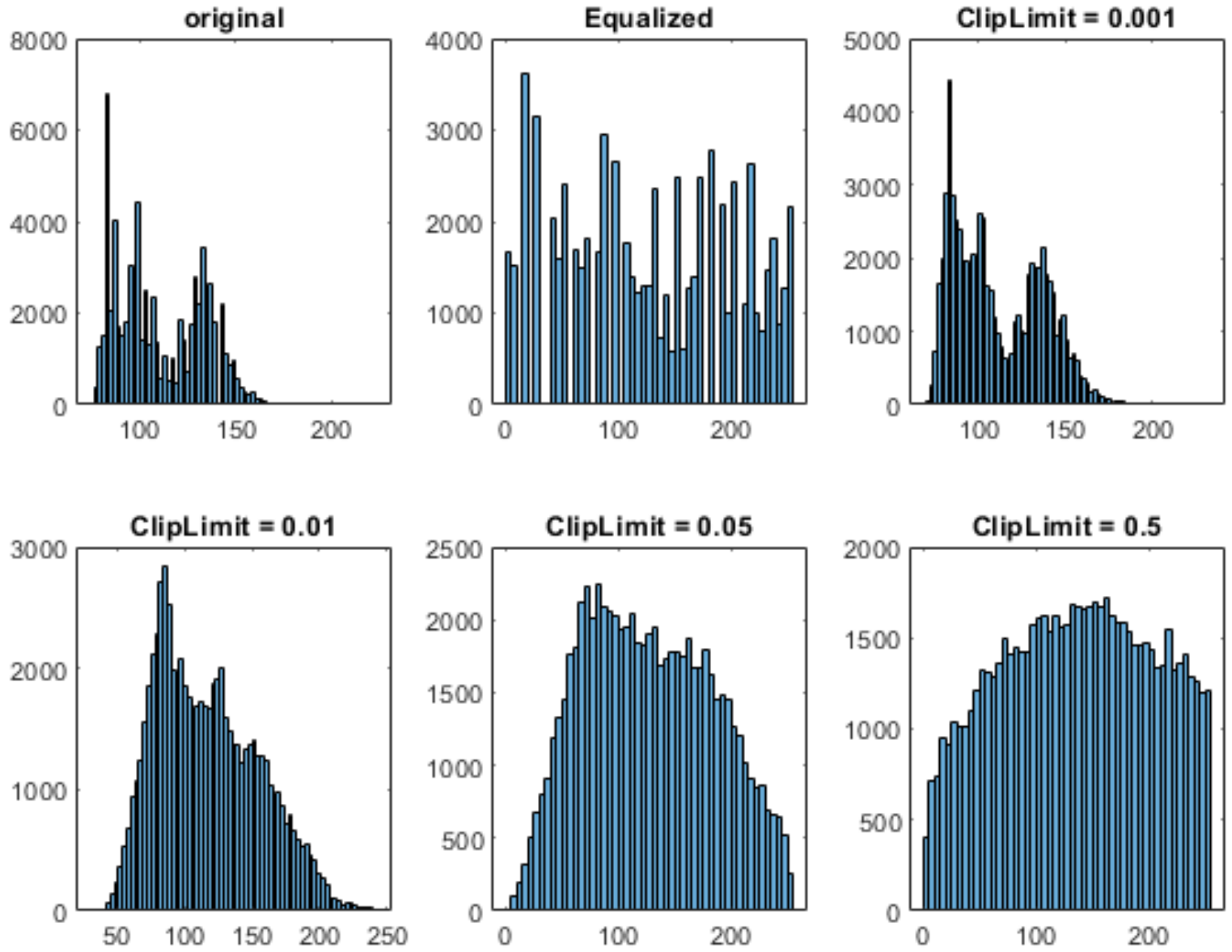
ClipLimit = 0.05



ClipLimit = 0.5



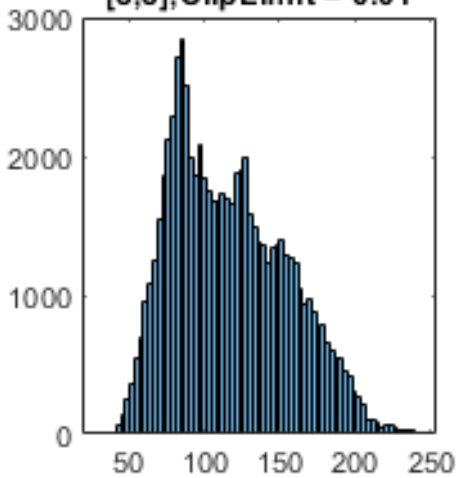
CLAHE Example:



CLAHE Example:

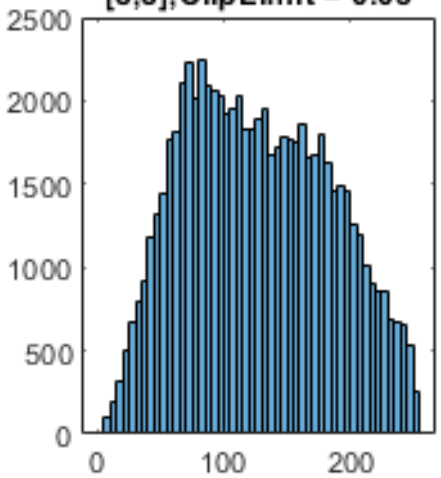
[8,8],ClipLimit = 0.01





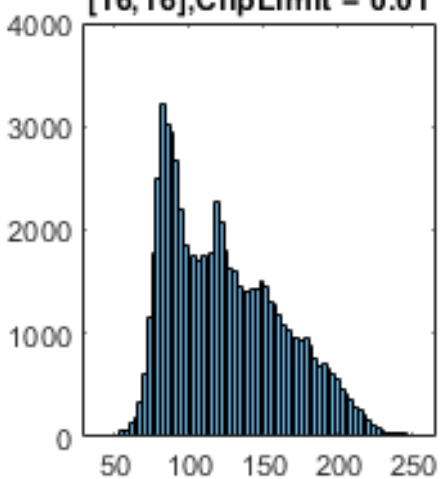
[8,8],ClipLimit = 0.05





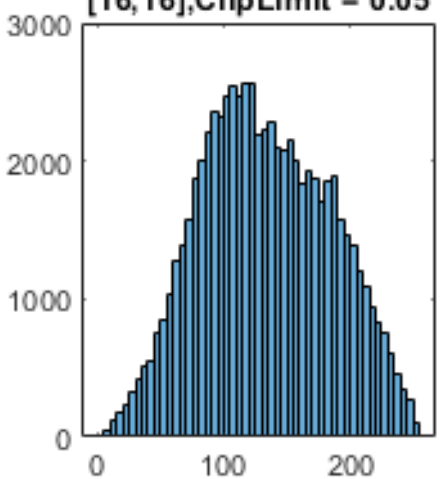
[16,16],ClipLimit = 0.01





[16,16],ClipLimit = 0.05





Other Equalization Methods:

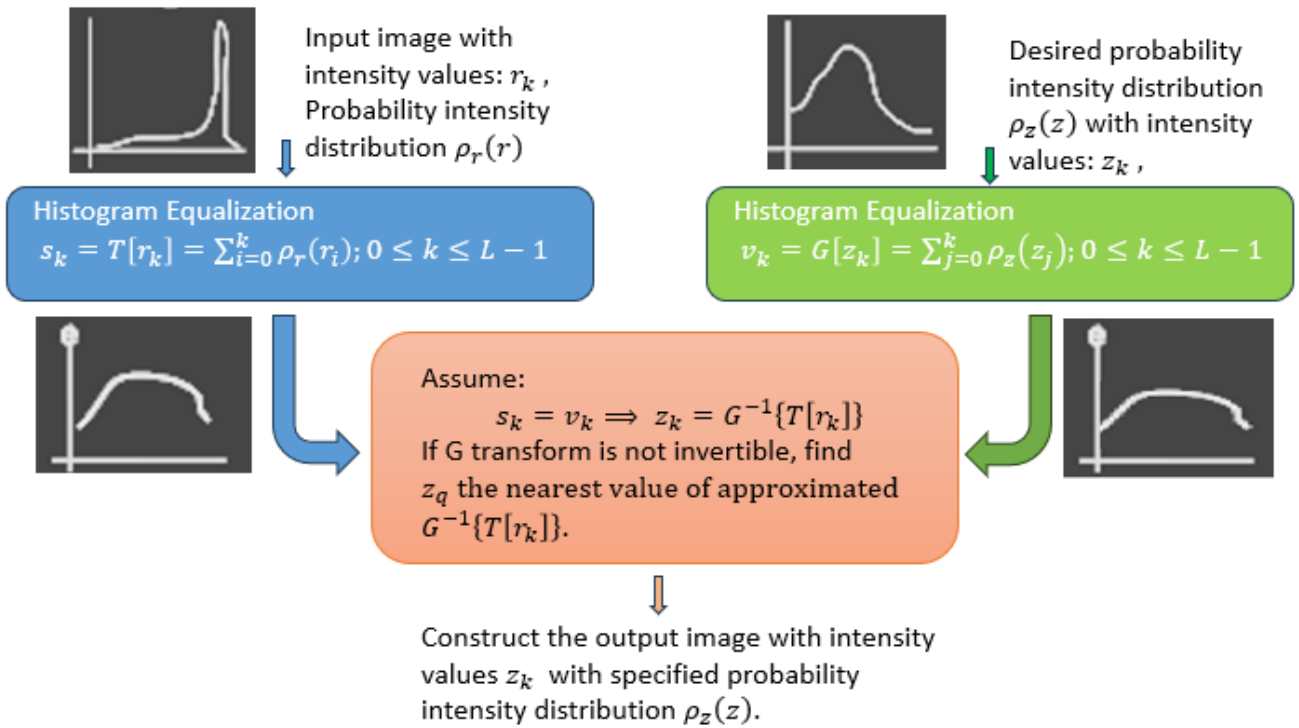
- Minimum mean brightness error bi-histogram equalization in contrast enhancement (MMBEBHE) (2003).
- Dualistic Sub-Image Histogram Equalization (DSIHE) 2013.
- Recursive Mean-Separate Histogram Equalization (RMSHE) 2021.

Learned Methods:

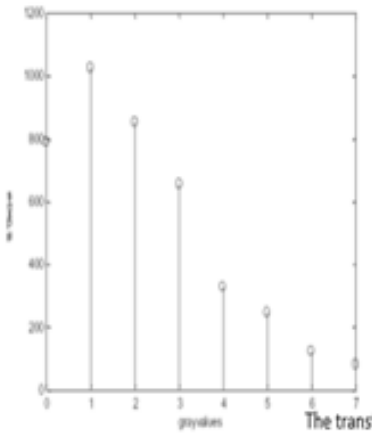
- Image-Adaptive CLAHE, in which a learned estimator selects the clip limit for individual tiles rather than using one fixed value throughout the image.
- Preventing Over Enhancement: Modified ICSO: an optimization method (2025).

Histogram Matching: (Histogram Specification)

Construct image with a prespecified histogram.
Suppose, the input image has probability density $\rho_r(r)$. We want to find a transformation $z = H(r)$, such that the probability density of the new intensity of image obtained by this transformation is a specified pdf $\rho_z(z)$.

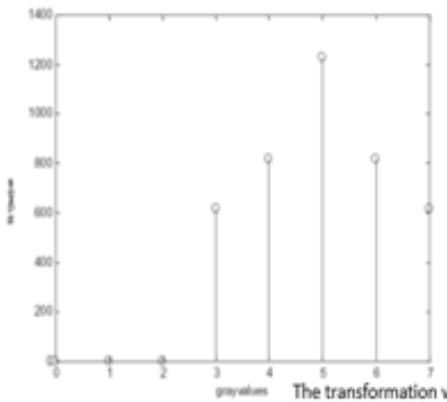


Histogram Matching: Numerical Example



k	r_k	n_k	$p(r_k)=n_k/n$	s_k
0	0	790	0.19	0.19
1	1/7	1023	0.25	0.44
2	2/7	850	0.21	0.65
3	3/7	656	0.16	0.81
4	4/7	329	0.08	0.89
5	5/7	245	0.06	0.95
6	6/7	122	0.03	0.98
7	1	81	0.02	1

The transformation $s=T(r)$ was obtained on r_k to forward uniform probability.



k	z_k	$p_{out}(z_k)$	v_k
0	0	0.00	0
1	1/7	0.00	0
2	2/7	0.00	0
3	3/7	0.15	0.15
4	4/7	0.20	0.35
5	5/7	0.30	0.65
6	6/7	0.20	0.85
7	1	0.15	1

The transformation $v=G(z)$ was Obtained on z_k to forward uniform probability

r_k	s_k	v_k	z_k	n_k	
0	0.19	0.15	3/7=0.43	790	790
1/7=0.14	0.44	0.35	4/7=0.57	1023	1023
2/7=0.29	0.65	0.65	5/7=0.71	850	850
3/7=0.43	0.81	0.85	6/7=0.86	656	985
4/7=0.57	0.89	0.85	6/7=0.86	329	
5/7=0.71	0.95	1	1	245	448
6/7=0.86	0.98	1	1	122	
1	1	1	1	81	

Histogram Matching: Example 1






Histogram Matching: Example 2

